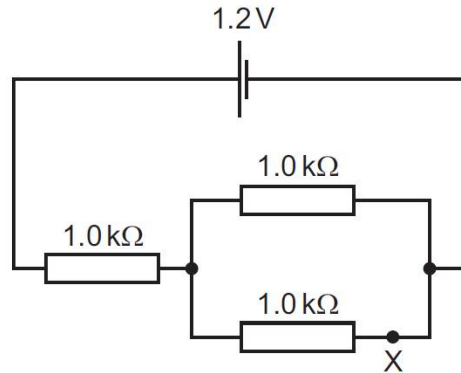


- 20 In the circuit below, 3 identical resistors of resistance $1.0 \text{ k}\Omega$ are connected to a cell of 1.2 V with negligible internal resistance as shown.



How many electrons pass through point X in a minute?

- A** 2.5×10^{15} **B** 1.5×10^{17} **C** 2.5×10^{18} **D** 1.5×10^{20}

Ans: B

$$\begin{aligned} V_{\text{parallel}} &= \frac{0.5}{1.5} (1.2) = 0.4 \text{ V} \\ I &= \frac{0.4}{1000} = 0.0004 \text{ A} \\ \frac{N_e}{t} &= \frac{0.0004}{1.6 \times 10^{-19}} = 2.5 \times 10^{15} \\ N_e &= 2.5 \times 10^{15} \times 60 = 1.5 \times 10^{17} \end{aligned}$$

- 21 Electrical sockets in a house are connected to a circuit called a ring main. The circuit is